

Layer 1: Actors



Designer/
Content designer



CI / PR / Nightly



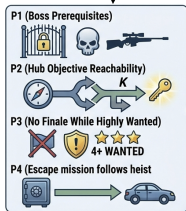
QA Engineer

Layer 2: Game specifics



Game content DB

Extract



Game rules (DSL)



Exported Story Graph

$G = (\mathcal{S}, \mathcal{A}, \mathcal{Q}, T, s_0, F)$

(States, Assets, Quests, Transitions,
initial and Final states)

Layer 3: Analysis Core

DSL compiler



Continuity Analysis
Engine (e.g., nuXmv)
Verify model against CTL

- hard locks
- soft locks
- disposable-critical
- shortcuts

detected issues

Layer 4: QA Triage and quality gate

content update/
rules revision



QA Triage

Store

Defect Report
Repository



Defect type and
Repair Suggestions



Abstract traces and
parameters
(seeds, context, etc.)

Re-create test
conditions and replay
selected traces



Instrumented Game Build

Selective replay request (validate fixes)